

Anj Asghar

Lead Senior Data Engineer | Data Architect | Cloud Data & Analytics Expert | Data Scientist | ML Engineer | Data Analyst

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SUMMARY

Principal Data Architect, Senior Data Engineer, and Data Scientist with 12+ years of experience building scalable, cloud native data platforms, real time streaming pipelines, and enterprise data architectures. Skilled in Data Engineering, Data Analytics, Data Science, Machine Learning, and Cloud Infrastructure, with 10+ years delivering advanced ML models, predictive analytics, and actionable business insights. Experienced in designing lakehouse architectures, ETL pipelines, real time event driven processing, and analytics dashboards. Strong background in AWS, GCP, Azure, Spark, Kafka, Flink, Airflow, and enterprise governance frameworks. Proven ability to lead cross functional teams, implement ML pipelines in production, and deliver business critical insights for data driven decision making.

Skills

Programming & Scripting

Python (Pandas, PySpark, NumPy, SciPy, Scikit Learn, FastAPI), SQL (T-SQL, PL/pgSQL, ANSI), Scala, Java, R, Bash, Shell, PyTorch, TensorFlow, Keras

Distributed Data Processing & Streaming

Distributed Data Processing & Streaming: Apache Spark (Structured Streaming), Apache Kafka (Schema Registry, Avro), Apache Flink, Apache Beam, Airflow (XComs, Sensors), Hadoop, HDFS

ML & Analytics

Feature Engineering, Regression, Classification, Time Series Forecasting, Recommendation Engines, NLP, Computer Vision, Model Deployment (Batch & Real Time), ML Pipelines, Model Monitoring, Tableau, Power BI, Looker, Superset, KPI Dashboards, A/B Testing

DevOps & Infrastructure

Terraform, Kubernetes (Helm, StatefulSets), Docker, Jenkins, GitOps, CI/CD, Serverless Computing, AWS Lambda

Data Storage & Modeling

Data Storage & Modeling: Delta Lake, Apache Hudi, Iceberg, Snowflake, Cassandra, MongoDB, Hive, HBase, PostgreSQL, MySQL, Redshift, BigQuery, Azure Synapse

Cloud Platforms & Lakehouse

AWS (Glue, Redshift, Lambda, S3, EMR, Athena), GCP (BigQuery, Dataflow, Pub/Sub, Composer), Azure (Synapse, Data Lake Gen2, Data Factory, Functions)

Data Governance & Security

RBAC, ABAC, Data Mesh, Metadata Lineage, OpenLineage, CDC, GDPR, HIPAA Compliance

Observability & Monitoring

Prometheus, Grafana, ELK Stack, DataDog, New Relic, Great Expectations

Professional Experience

Lead Senior Data Engineer & Data Scientist, Zaloni

02/2024 – Present

- Designed and implemented cloud native ETL pipelines using AWS Glue, Redshift, and Lambda for real time data integration.
- Built high volume streaming pipelines with Kafka, Flink, and Spark achieving sub second latency.
- Implemented metadata lineage, data governance, and compliance using Apache Atlas and AWS Glue Data Catalog.

- Automated deployments using Terraform and Jenkins CI/CD, improving reliability and reducing errors.
- Optimized infrastructure costs via serverless computing, auto scaling, and resource tuning.
- Mentored engineering teams on modular pipeline design, architecture best practices, and performance optimization.
- Developed ML pipelines for predictive analytics, including regression, classification, time series forecasting, and recommendation engines; deployed models to production.
- Created interactive dashboards and KPI reports in Tableau and Power BI for multiple business units.
- Applied A/B testing and analytics experiments to measure business impact of ML models.

Senior Data Platform Engineer & Data Scientist, Sigma Computing

06/2022 – 01/2024

- Designed lakehouse architecture using AWS S3, Redshift, and Delta Lake for structured and semi structured data.
- Built event driven real time pipelines with Kafka and Flink.
- Developed ML models for forecasting, churn prediction, and recommendation systems; deployed pipelines using Spark and Airflow.
- Implemented data quality frameworks with Great Expectations for batch and streaming pipelines.
- Optimized Redshift and BigQuery performance via partitioning, clustering, and tuning.
- Enforced RBAC policies and secured sensitive data across environments.
- Delivered business dashboards, KPIs, and analytics insights supporting decision making.

Data Engineer, Starschema

09/2018 – 05/2022

- Built Python and SQL based ETL pipelines processing 2+ TB daily, reducing runtime by hours.
- Designed scalable relational database models in PostgreSQL and MySQL.
- Implemented CI/CD pipelines with Debezium (less than 5-sec replication latency).
- Built ML feature engineering pipelines to support predictive modeling and analytics.
- Developed Power BI and Tableau dashboards for 12+ business units, driving operational insights

Data Engineering Associate, Altera Data

04/2013 – 08/2018

- Developed large scale ETL pipelines using Apache Airflow and Python.
- Consolidated data from multiple operational systems into a centralized cloud warehouse.
- Built real time feature engineering pipelines supporting ML model training and predictions.
- Automated data validation and quality checks with Great Expectations.
- Implemented monitoring using Prometheus and Grafana.
- Reduced cloud costs with serverless computing and storage tiering strategies.

Education

Bachelor of Computer Science, University of Punjab

2008 – 2012